

Pathology Foundation Models are Scanner Sensitive

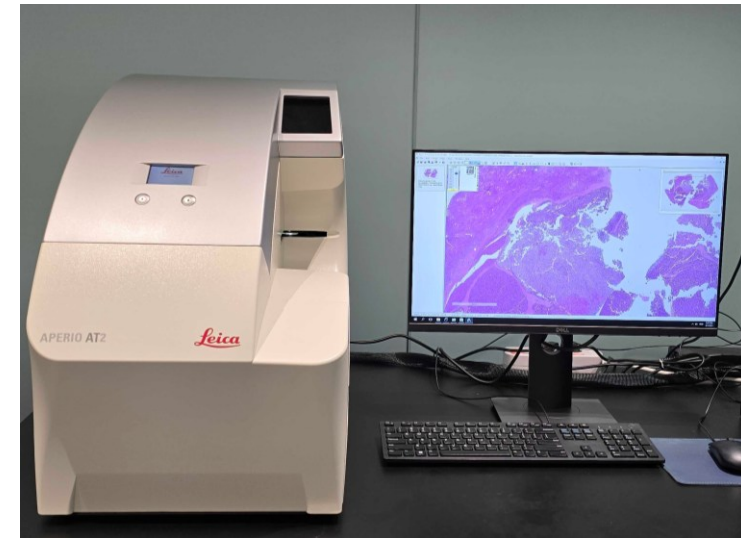
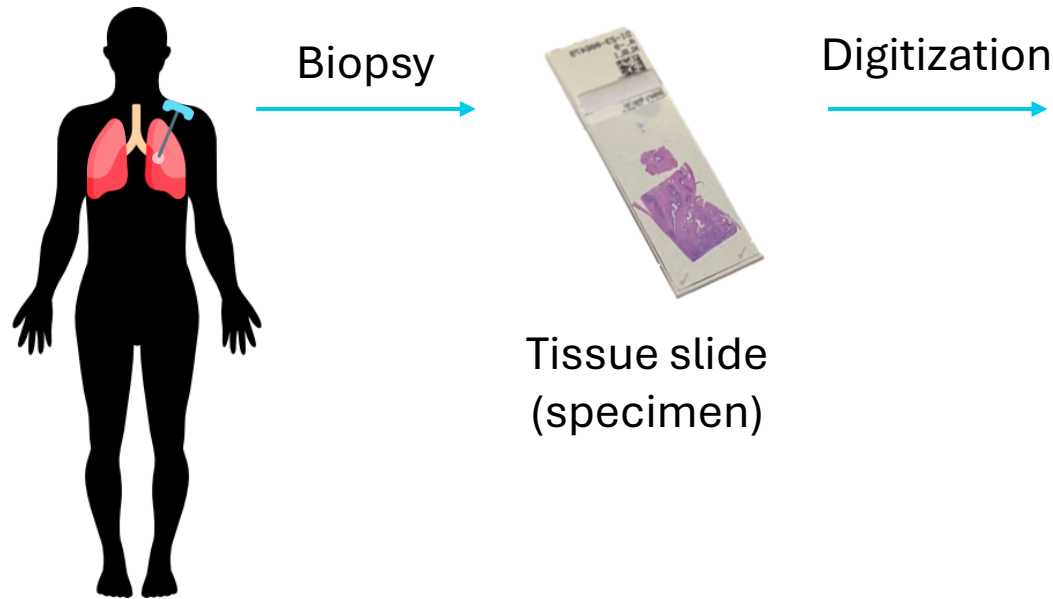
Benchmark and Mitigation with Contrastive ScanGen Loss

Gianluca Carloni*, Biagio Brattoli*, Seongho Keum, Jongchan Park, Taebum Lee, Changho Ahn,
and **Sérgio Pereira**

Background and Motivation

A Typical Pipeline

- Digital pathology requires the digitization of **tissue slides** into Whole Slide Images (**WSIs**) using **slide scanners**.



Background and Motivation

The Effect of Different Setups

- Slide scanners capture the underlying tissue accurately.
- **Different devices** may result in **different** image **characteristics**.

- Optical system → image sharpness



- Light source → contrast, artifacts



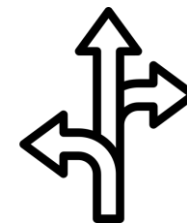
- Software → data compression



Background and Motivation

At Global Scale

- Different scanners are used **worldwide**.
 - Leica and 3DHistech in Europe.
 - Leica and Philips in America.
 - Hamamatsu and Philips in Asia.
- Any performance issues associated with a particular scanner could lead to **regional disparities** in patient care.



Background and Motivation

The Foundation Model Era

- The adoption of computational pathology (**CPath**) foundation models (**FMs**) is **growing**.
- They **promise** better domain generalization capabilities.
- Still, they may **encode** several **biases** (hosp. envs, slide identity).

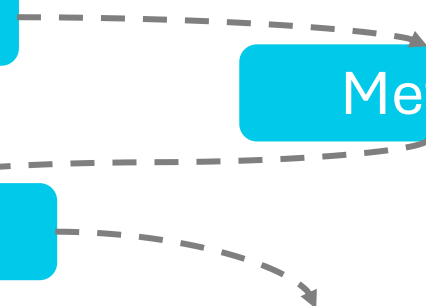
Do CPath FMs **encode bias from** the digital **scanners** used for reading tissue specimens?

Benchmark

Metric

Loss

Relevance

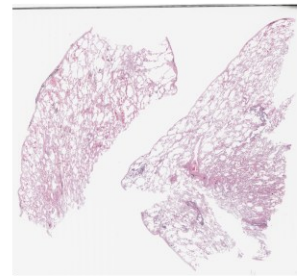


Methods

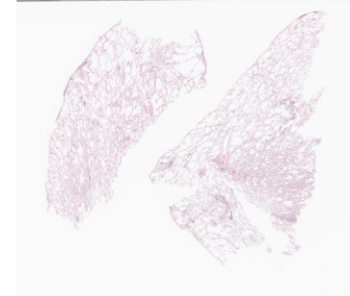
Building a Benchmark

- We collected a **new dataset** to **benchmark** state-of-the-art CPath FMs.
- Images of the same specimen captured from **6 scanners** at 40x magnification
 - Plus, one scanner at 20x
- $323 \times 7 = \mathbf{2261}$ **WSIs**
- Annotated for **EGFR mutation**

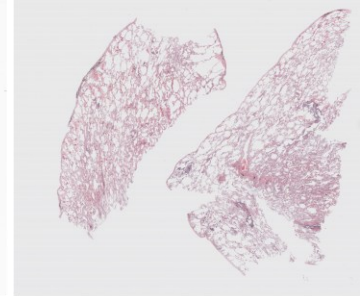
Leica Aperio AT2



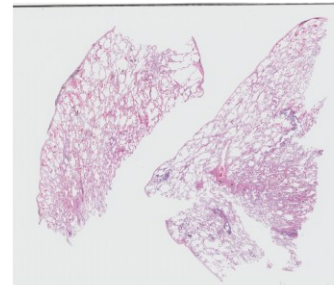
Leica Aperio GT450



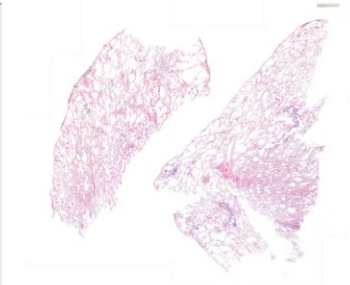
Roche V DP200



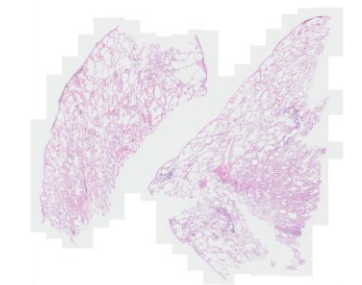
Hamamatsu S360 MD



3DHistech P1000



Philips UFS B300



Methods

What is Scanner Bias?

Quantitative Viewpoint

- Model output changes based on which scanner was used, despite the tissue content being the same
- Define the **Coefficient of Variation** (CoV)
 - The lower, the better!

$$\text{CoV} = \frac{1}{M} \sum_{i=1}^M \left(\frac{\sigma(\hat{y}_i^{s_1}, \hat{y}_i^{s_2}, \dots, \hat{y}_i^{s_S})}{|\mu(\hat{y}_i^{s_1}, \hat{y}_i^{s_2}, \dots, \hat{y}_i^{s_S})|} \right)$$

Num. of specimens → M
Standard deviation → $\sigma(\hat{y}_i^{s_1}, \hat{y}_i^{s_2}, \dots, \hat{y}_i^{s_S})$
Num. of scanners → S
Mean → $|\mu(\hat{y}_i^{s_1}, \hat{y}_i^{s_2}, \dots, \hat{y}_i^{s_S})|$

A typical scenario:

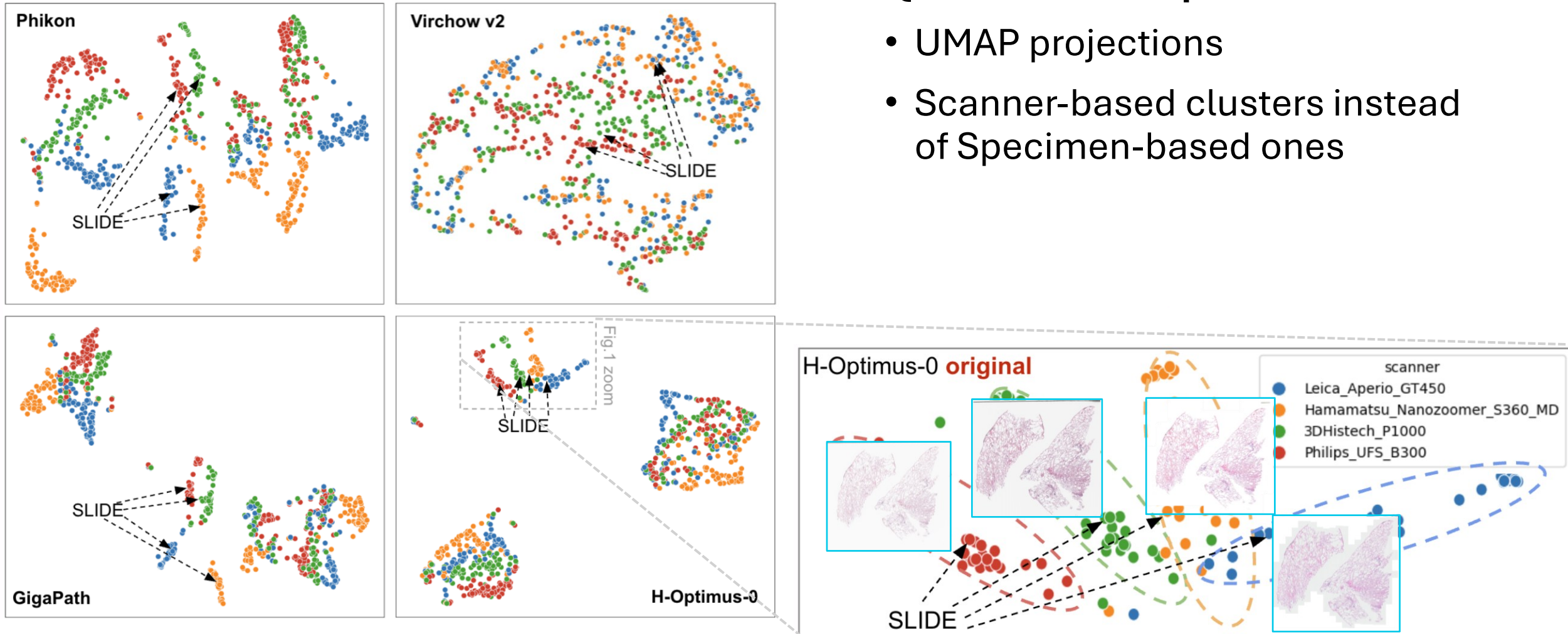
Specimen_id	Ground Truth	3DHistech P1000	Hamamatsu N S360MD	Leica_Aperio_GT450	Leica_aperio_AT2	Philips_UFS_B300	Roche_V_DP200	CoV
S123456	0	0.198	-0.267	0.275	-0.139	-0.127	0.065	234.798
S123485	0	0.209	-0.290	0.267	-0.252	-0.156	0.210	133.713
S123854	0	0.293	-0.316	-0.043	-0.112	0.425	-0.233	118.926
S128453	1	0.485	-0.235	0.173	-0.415	-0.088	0.101	92.747
S183452	0	0.121	-0.280	0.006	-0.340	-0.099	0.647	39.087
S823451	0	0.205	-0.256	-0.113	-0.190	0.061	0.247	27.365
S828450	1	0.501	-0.371	0.119	-0.261	-0.205	0.330	18.595
S883459	0	0.558	-0.421	-0.085	-0.317	0.132	0.000	15.764
S...	...	...	...	...	...	...	...	...

What we aim for:

S183451	1	1.792	2.002	2.070	2.064	1.669	2.256	0.107
S881156	0	-1.157	-0.951	-1.131	-1.228	-1.203	-1.191	0.088

Methods

What is Scanner Bias?

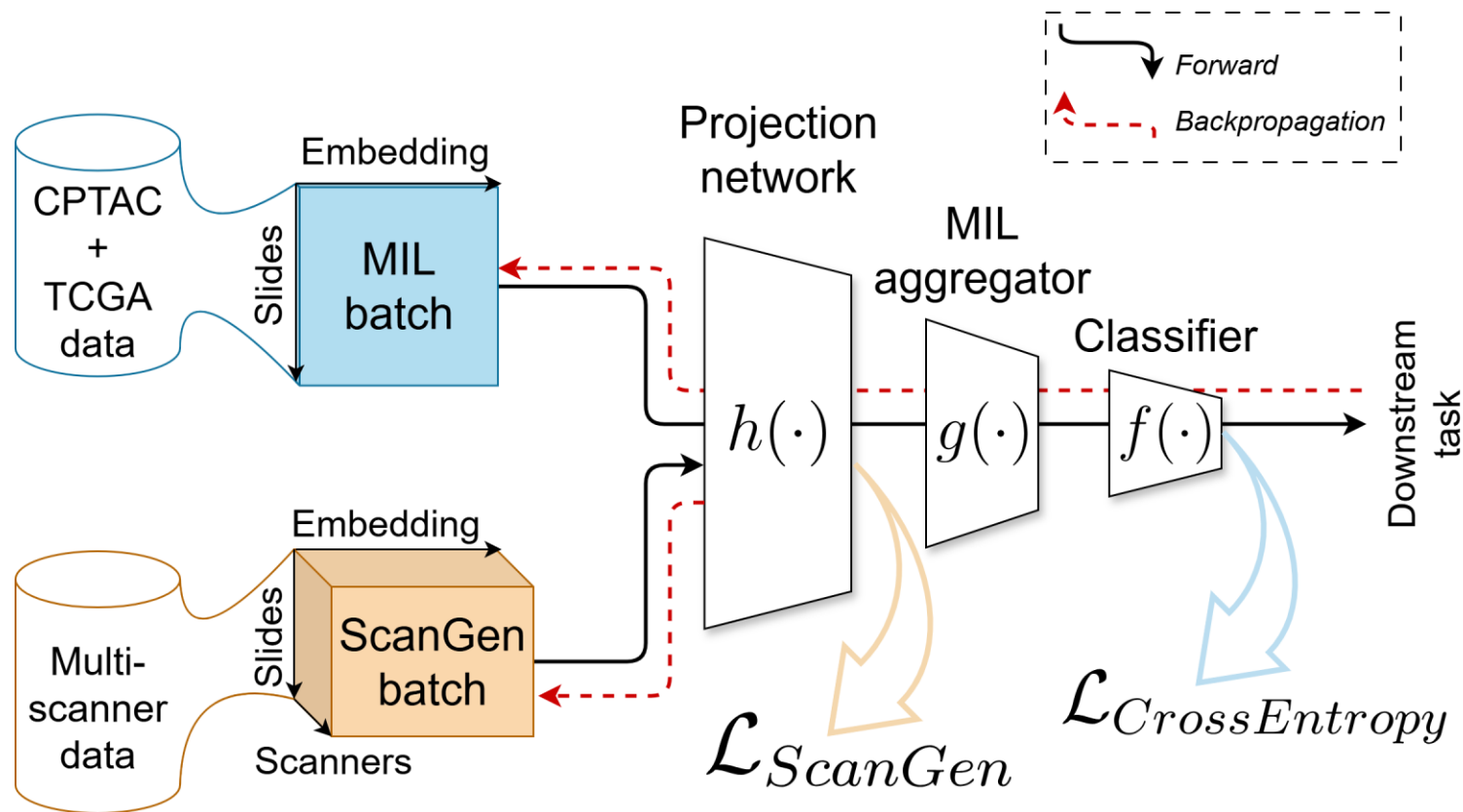


Qualitative Viewpoint

- UMAP projections
- Scanner-based clusters instead of Specimen-based ones

Methods

Framework and ScanGen Loss



- Shared model, two data flows
- Regular MIL batch from EGFR data
- Proj. > Aggr. > Class. cascade
- Logits for standard cross-entropy loss
- Batch of embeddings identified by slide and scanner are fed to Projection network

Methods

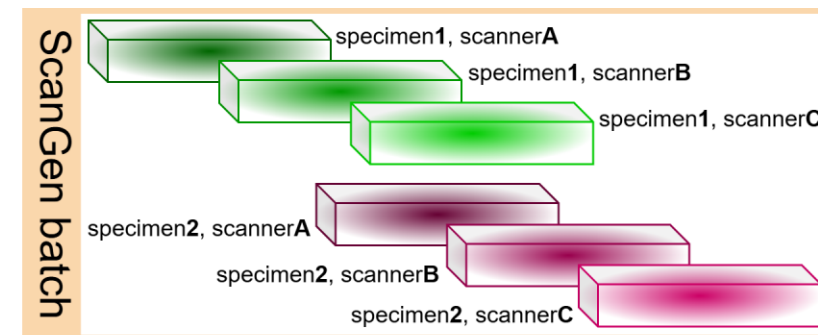
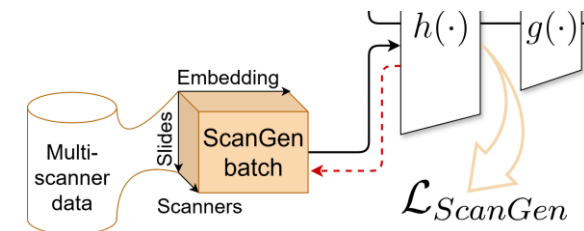
Framework and ScanGen Loss

- Two specimens: i and j
- Two scanners: s_1 and s_2

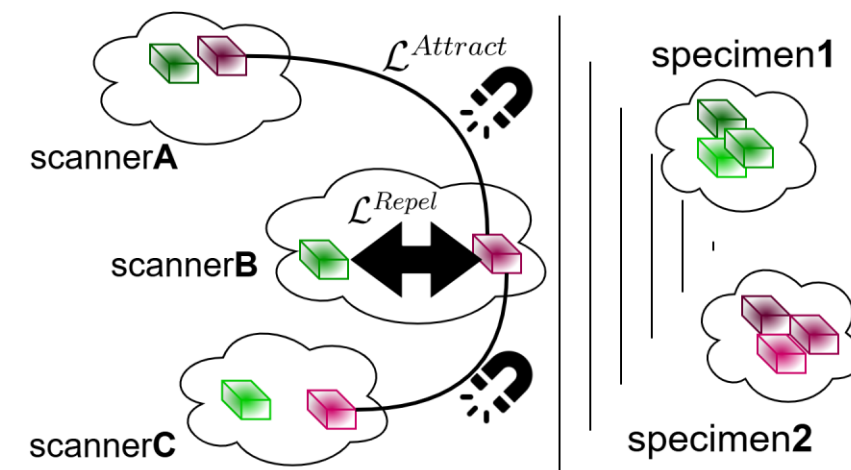
$$\mathcal{L}_{ScanGen} = (1 - \alpha) \underbrace{(d(h(x_i^{s_1}), h(x_i^{s_2})))^2}_{\mathcal{L}^{Attract}} + \underbrace{\alpha (\max(0, r - d(h(x_i^{s_1}), h(x_j^{s_1}))))^2}_{\mathcal{L}^{Repel}}$$

Annotations for the equation:

- $d(a, b) = 1 - \frac{a \cdot b}{|a| \cdot |b|}$ (Cosine distance)
- α (hyperparameter)
- r (margin)



$$\mathcal{L}_{ScanGen} = (1 - \alpha) \mathcal{L}^{Attract} + \alpha \mathcal{L}^{Repel}$$

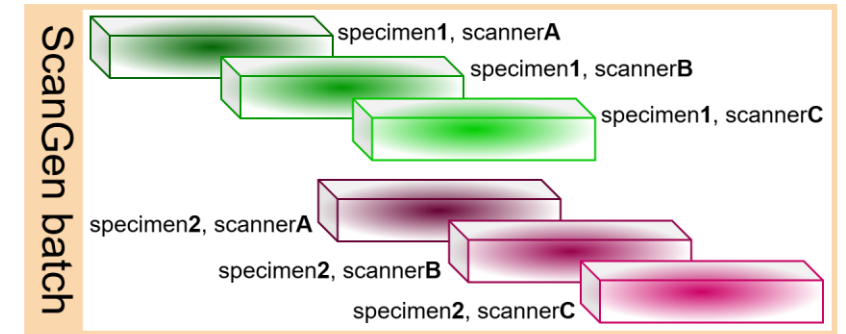
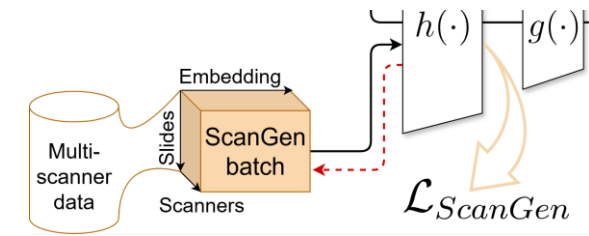


Methods

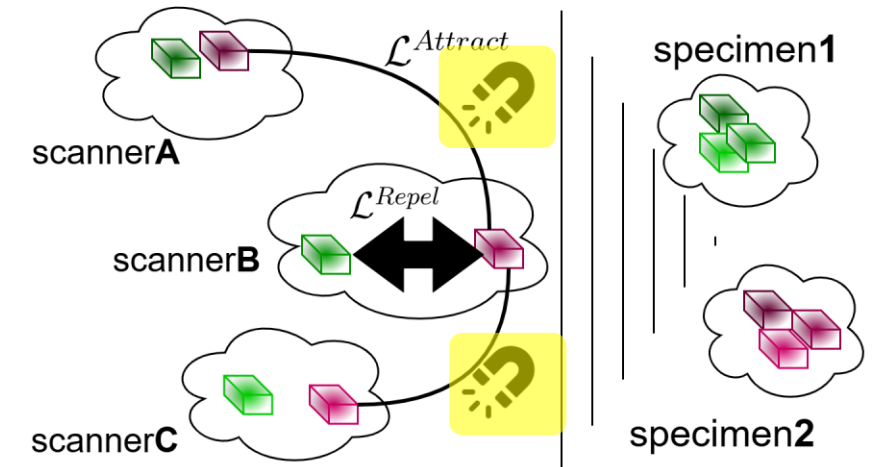
Framework and ScanGen Loss

$$\mathcal{L}_{ScanGen} = (1 - \alpha) \underbrace{(d(h(x_i^{s_1}), h(x_i^{s_2})))^2}_{\mathcal{L}^{Attract}} + \underbrace{\alpha (\max(0, r - d(h(x_i^{s_1}), h(x_j^{s_1}))))^2}_{\mathcal{L}^{Repel}}$$

- **Attraction:** encourage **same-specimen different-scanner** samples to be close
- **Repulsion:** push same-scanner different-specimen samples apart



$$\mathcal{L}_{ScanGen} = (1 - \alpha) \mathcal{L}^{Attract} + \alpha \mathcal{L}^{Repel}$$

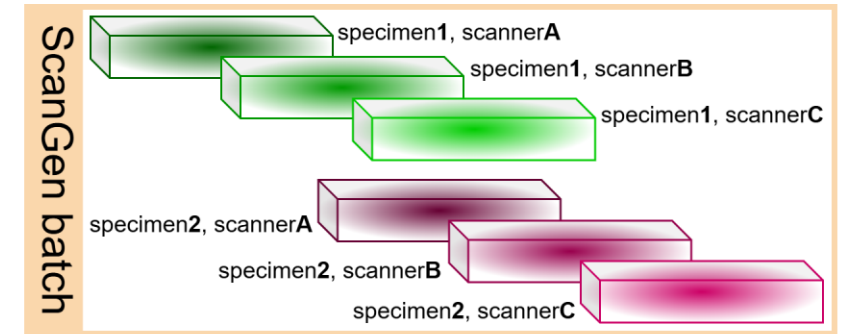
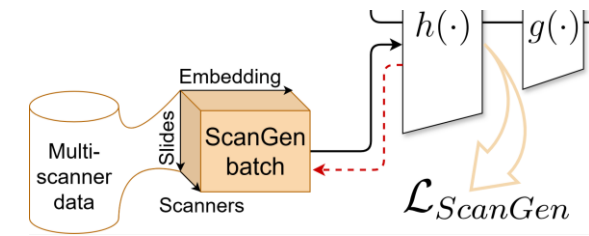


Methods

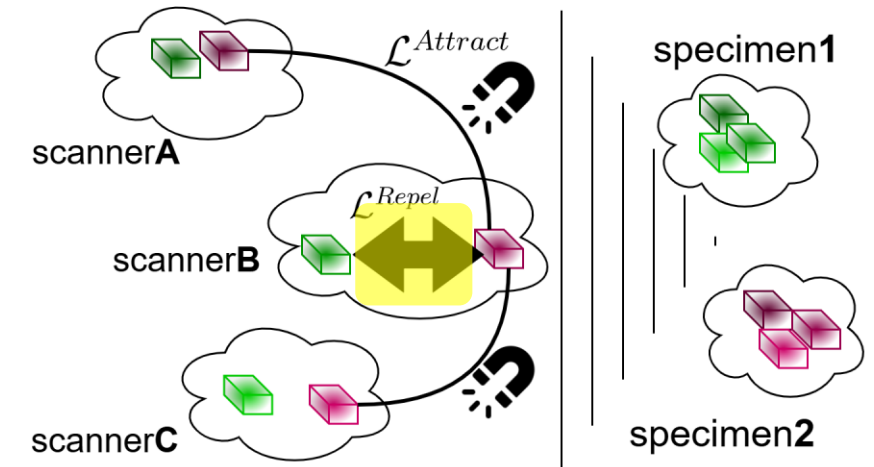
Framework and ScanGen Loss

$$\mathcal{L}_{ScanGen} = (1 - \alpha) \underbrace{(d(h(x_i^{s_1}), h(x_i^{s_2})))^2}_{\mathcal{L}^{Attract}} + \underbrace{\alpha (\max(0, r - d(h(x_i^{s_1}), h(x_j^{s_1}))))^2}_{\mathcal{L}^{Repel}}$$

- Attraction: encourage same-specimen different-scanner samples to be close
- **Repulsion**: push **same-scanner** different-specimen samples apart



$$\mathcal{L}_{ScanGen} = (1 - \alpha) \mathcal{L}^{Attract} + \alpha \mathcal{L}^{Repel}$$



Methods

Framework and ScanGen Loss

- Final loss: **downstream task** loss (EGFR cross-entropy) and **ScanGen** loss

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N (\mathcal{L}_{CE}(\hat{y}_i, y_i)) + \lambda \frac{1}{M} \sum_{i,j=1}^M \sum_{u,v=1}^S \mathcal{L}_{ScanGen}(x_i^{s_u}, x_i^{s_v}, x_j^{s_u}; h)_{i \neq j, u \neq v}$$

Batch size downstream task

hparam

Batch size ScanGen task

Num. of scanners

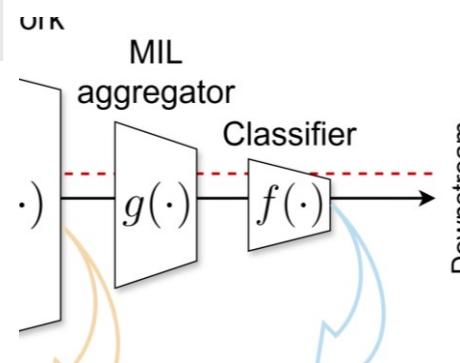
Aggregator-Classifiers

Global Average Pooling + MLP

AB-MIL [1]

DS-MIL [2]

Slot-MIL [3]



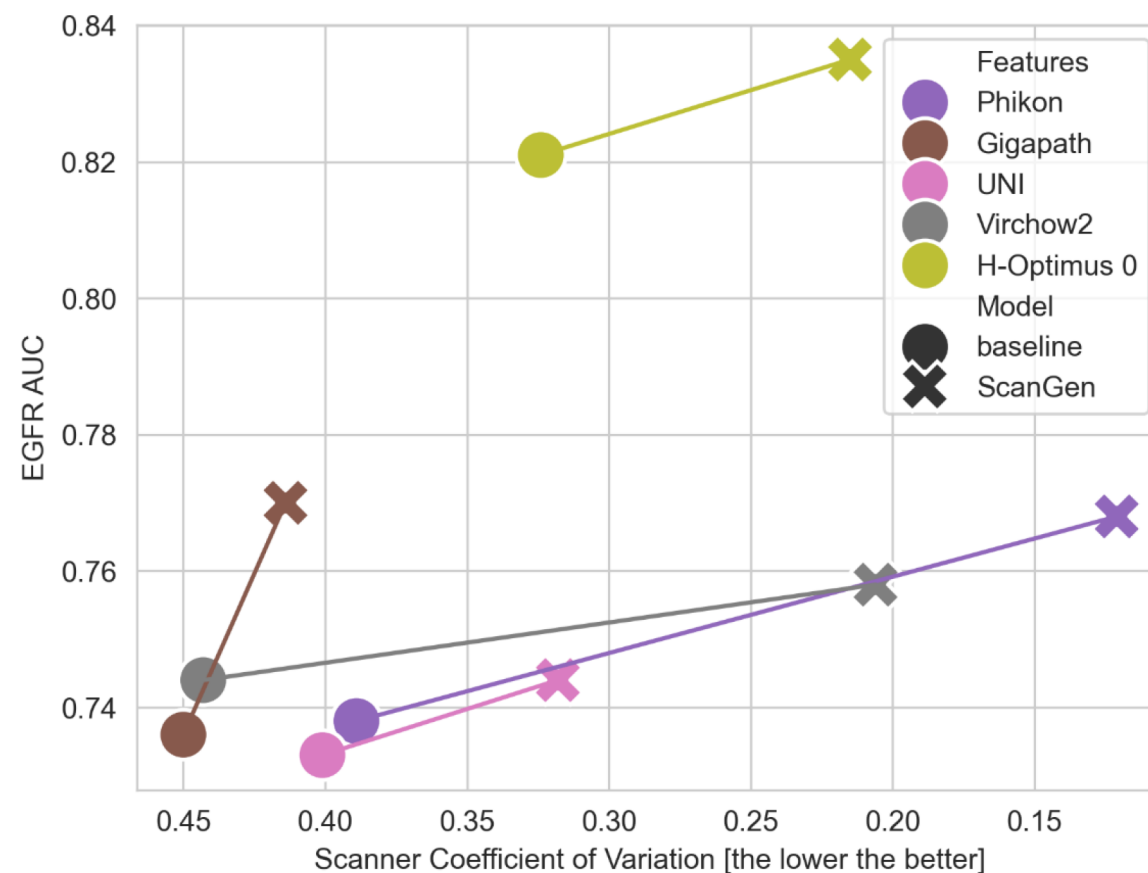
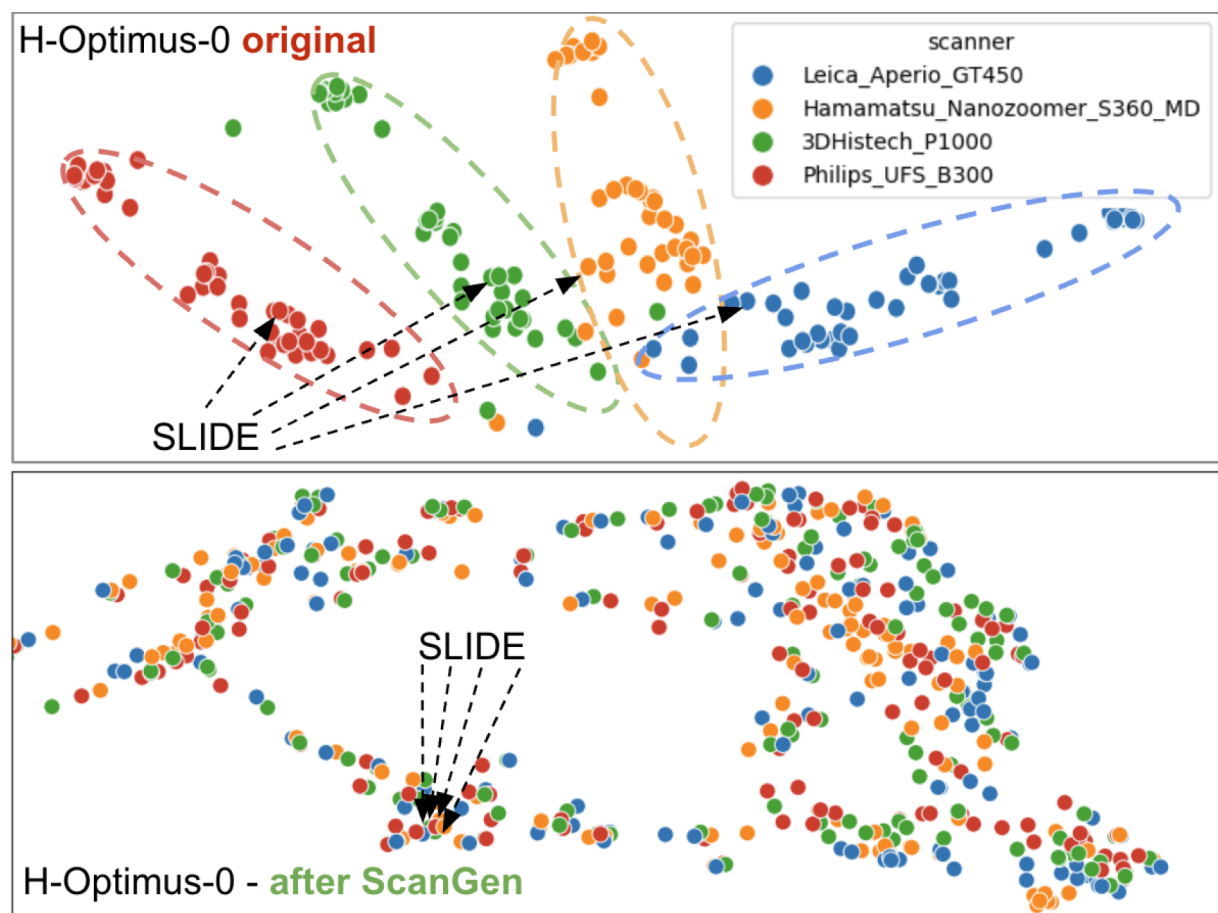
Hyperparameter

Appropriate domain

ScanGen repulsion	α	0.12 – 0.21
ScanGen margin	r	0.90 – 1.10
ScanGen contribution	λ	0.50 – 1.00

Results

- **Aligns** same-specimen **representations** regardless of scanner
- **Lowers** the scanner Coefficient of Variation (**CoV**)
- **Improves** the downstream task AUC



Results

GAP+MLP,
for different
FMs

Embedding	Use <i>ScanGen</i>	EGFR AUC [↑]	Scanner CoV [↓]	Magnification CoV [↓]
Phikon[6]	No	0.738	0.389	0.158
	Yes	0.768	0.121 -68,9%	0.053 -66,4%
UNI[3]	No	0.733	0.401	0.116
	Yes	0.744	0.318 -20,7%	0.089 -23,3%
Virchow[22]	No	0.744	0.443	0.125
	Yes	0.758	0.206 -53,5%	0.077 -38,4%
Gigapath[18]	No	0.736	0.450	0.174
	Yes	0.770	0.414 -8,00%	0.193 +10,9%
H-Optimus[15]	No	0.821	0.324	0.117
	Yes	0.835	0.215 -33,6%	0.108 -7,70%

the lower the better

MIL approach	GAP		AB-MIL[8]		DS-MIL[13]		SlotMIL[12]	
	AUC	CoV [↓]	AUC	CoV [↓]	AUC	CoV [↓]	AUC	CoV [↓]
EGFR	0.821	0.324	0.828	0.251	0.834	0.265	0.822	0.338
EGFR + <i>ScanGen</i>	0.835	0.215	0.835	0.224	0.828	0.214	0.826	0.227

H-Optimus0
features,
for different MIL
approaches

-33,6%

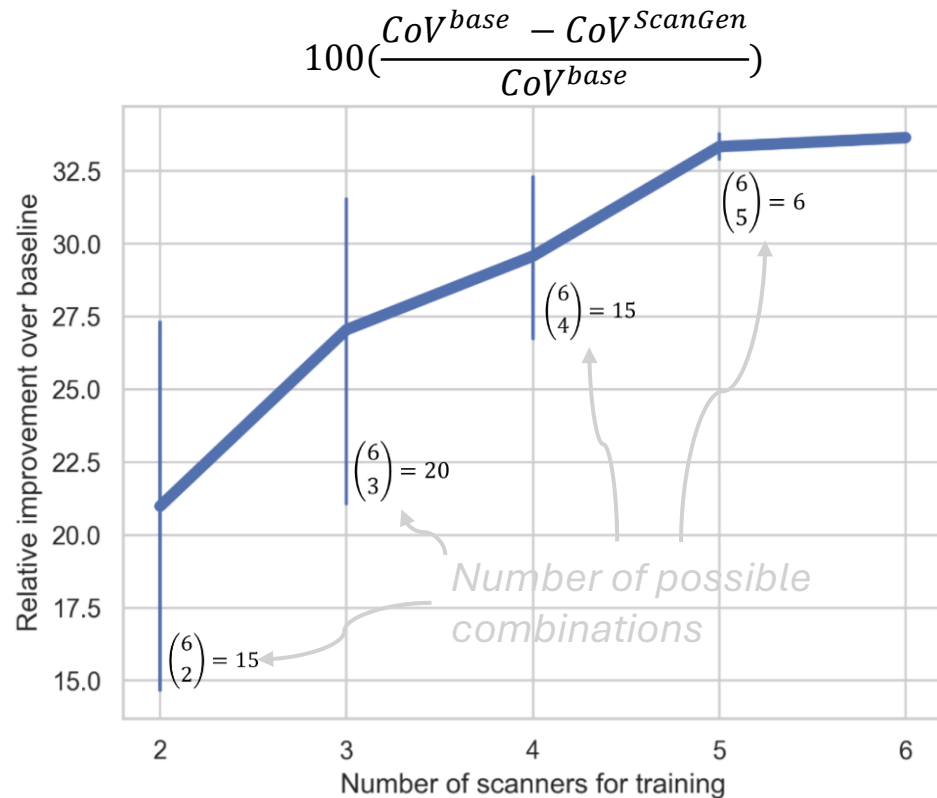
-10,7%

-19,2%

-32,8%

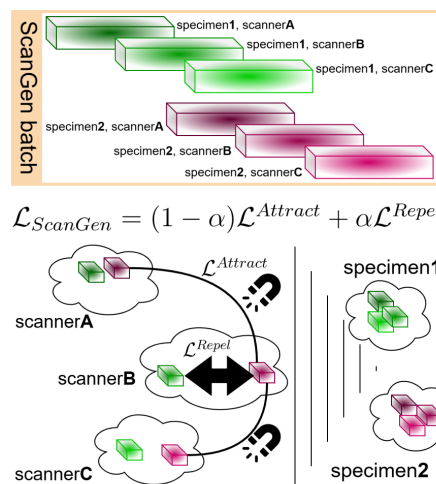
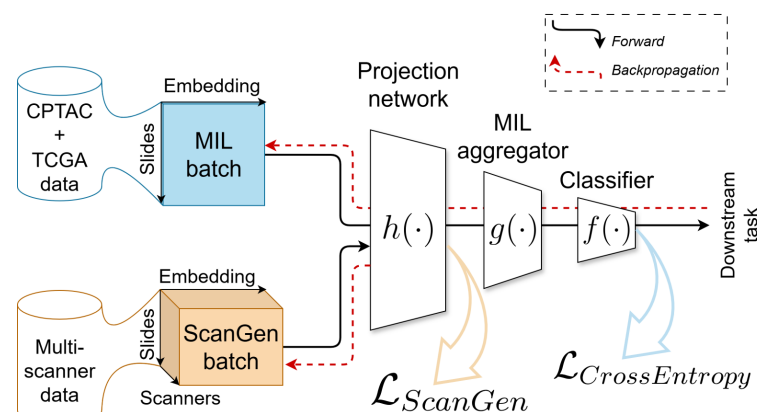
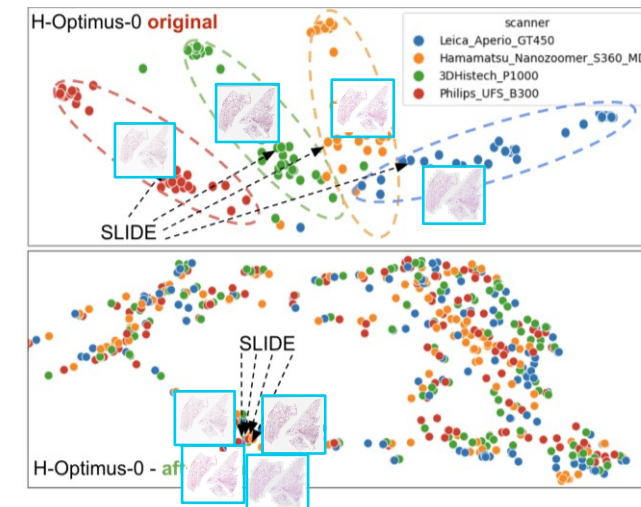
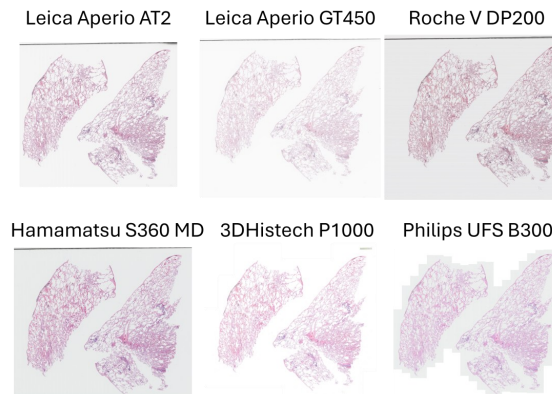
Ablation: ScanGen with Limited Scanners

- **Training:** different combinations of **2** to **6** scanner types
- **Testing:** on the same external set containing all **6** scanner sources

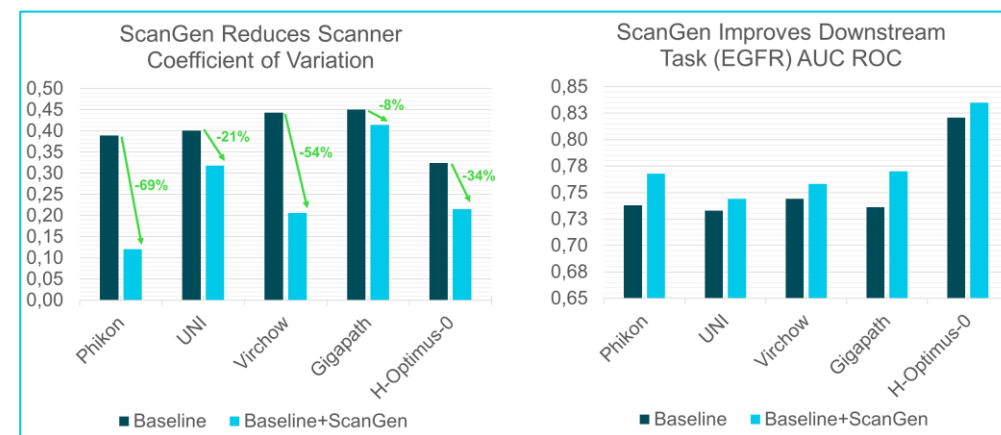


- When **more scanners** are used to train ScanGen, the **gain** in prediction agreement **increases**
- **Converges** at **5** scanners
- **3 scanners** are sufficient for **+28%**
- Using a **diverse** set of **scanners** leads to the **best generalization**; using **similar scanners** (e.g., Aperio AT2 and GT450) does **not** benefit **much**

Conclusion



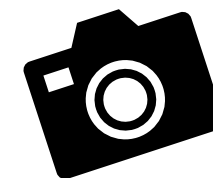
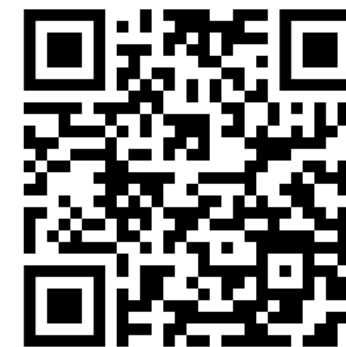
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Thank you

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Read our paper!

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